

Language Models and Agents (LMA)

Phase 1 Report

Data Collection & Preprocessing for Telugu and Bhojpuri

Kspsvln Siddardha Kumar Kavuri

Roll Number 2025201061

kspsvlnsiddardha.k@students.iiit.ac.in

August 19, 2026

Phase 1 Completion Date: August 19, 2026

Dataset Available on Kaggle

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>

Contents

1	Phase 1 Data Report	3
1.1	Data Collection Summary	3
1.1.1	Telugu (Model H)	3
1.1.2	Bhojpuri (Model L)	3
1.2	Data Processing Pipeline	3
1.2.1	Cleaning Process	3
1.2.2	Data Splits (80/10/10)	4
2	Data Collection Overview	4
2.1	Methodology	4
2.1.1	Telugu (Model H) - Higher-Resource Language	4
2.1.2	Bhojpuri (Model L) - Lower-Resource Language	4
2.2	Data Sources	4
3	Data Cleaning Pipeline	5
3.1	7-Stage Cleaning Process	5
3.2	Quality Metrics	5
3.2.1	Telugu Cleaning Analysis	5
3.2.2	Bhojpuri Cleaning Analysis	5
4	Data Statistics & Visualizations	6
4.1	Collection Metrics Comparison	6
4.2	Cleaning Pipeline Results	6
4.3	Train/Validation/Test Split Distribution	7
4.4	Data Source Composition (Bhojpuri)	7

5	Train/Validation/Test Splits	7
5.1	Split Strategy	7
5.2	Telugu Splits	8
5.3	Bhojpuri Splits	8
6	Detailed Statistics	8
6.1	Telugu (Model H) Statistics	8
6.2	Bhojpuri (Model L) Statistics	9
7	Technical Implementation	9
7.1	Data Pipeline Architecture	9
7.2	File Structure	9
8	Quality Assurance	10
8.1	Validation Checklist	10
8.2	Reproducibility	10
9	Challenges & Solutions	11
9.1	Challenge 1: Bhojpuri Data Scarcity	11
9.2	Challenge 2: OCR Quality	11
9.3	Challenge 3: Scale Management	11
10	Dataset Repository	11
10.1	Dataset Contents	11
A	Configuration Files	12
A.1	Telugu config.json Structure	12
B	References	12

1 Phase 1 Data Report

This report provides a comprehensive data collection and preprocessing report for two monolingual Transformer language models:

- **Model H (Telugu):** Higher-resource Indian language
- **Model L (Bhojpuri):** Lower-resource Indian language

1.1 Data Collection Summary

1.1.1 Telugu (Model H)

- Total lines collected: 25,349,199
- Estimated tokens: 4,421,917,647 (4.4B)
- Web-scraped texts: 29,206 (89.7% pass rate from 32,549 raw)
- Manual corpus: 21,458,261 lines (te.txt)
- OCR augmentation: 3,766,703 lines
- Target: 500M tokens | Progress: 884.4%

1.1.2 Bhojpuri (Model L)

- Total lines collected: 776,801
- Estimated tokens: 136,606,695 (136.6M)
- Web-scraped texts: 3,234 (97.2% pass rate from 3,327 raw)
- HuggingFace Corpus: 386,000 documents (25M tokens)
- OCR augmentation: 511,599 lines
- Machine Translation (Hindi→Bhojpuri): 15,852 lines
- Target: 500M tokens | Progress: 27.3%

1.2 Data Processing Pipeline

1.2.1 Cleaning Process

- Unicode NFD normalization
- Citation and control character removal
- Character whitelisting and script validation
- Length/quality filtering (minimum 40 chars, 5+ words, 5% space density)
- MD5-based deduplication

1.2.2 Data Splits (80/10/10)

- Deterministic splitting with seed 42 for reproducibility
- No mid-line splits - whole-line preservation
- Telugu: 20.3M train / 2.5M val / 2.5M test
- Bhojpuri: 627.9K train / 78.1K val / 78.4K test

2 Data Collection Overview

2.1 Methodology

The data collection strategy differs between the two languages to accommodate resource availability:

2.1.1 Telugu (Model H) - Higher-Resource Language

- **Primary Source:** Existing corpus (te.txt) - 21.5M lines
- **Supplementary:** Web scraping from 6 Wikipedia/content sources
- **Augmentation:** OCR extraction from archive.org texts
- **Total:** 25.3M lines training data

2.1.2 Bhojpuri (Model L) - Lower-Resource Language

- **Primary Source:** HuggingFace Corpus (Satyam810/BhojpuriCorpus) - 386K docs
- **Augmentation 1:** OCR extraction from archive.org Hindi texts - 511.6K lines
- **Augmentation 2:** Web scraping from Hindi news sites (Devanagari proxy) - 3.2K lines
- **Augmentation 3:** Machine Translation (Hindi→Bhojpuri via NLLB-200) - 15.8K lines
- **Total:** 776.8K lines training data (27.3% toward 500M token target)

2.2 Data Sources

Language	Source	Contribution
Telugu	te.txt (existing)	21.5M lines (99.9%)
	te.wikipedia.org	3K articles
	te.wikibooks.org	500+ pages
	te.wikiquote.org	300+ quotes
	News portals	eenadu.net, greatandhra.com
	archive.org OCR	3.8M lines (augmentation)
Bhojpuri	HF Corpus	386K docs (~25M tokens)
	archive.org OCR	511.6K lines
	bho.wikipedia.org	~8.9K articles
	Hindi News	Via NLLB-200 MT
	Web Scraping	3.2K lines (Hindi proxy)
	GlottCC-V1	949 docs
	HPLT 3.0	~32.8K docs
	fish-food	262K rows

Table 1: Data sources for both languages

3 Data Cleaning Pipeline

3.1 7-Stage Cleaning Process

All data undergoes a rigorous 7-stage cleaning pipeline:

1. **Unicode Normalization:** Convert to NFD (Canonical Decomposition)
2. **Citation Removal:** Remove Wikipedia citations and references
3. **Control Character Removal:** Strip non-printable characters
4. **Character Whitelisting:** Retain only valid Unicode ranges
5. **Script Validation:** Language-specific script filtering
6. **Length/Quality Filters:** Minimum length, word count, density checks
7. **Deduplication:** MD5-based exact duplicate removal

3.2 Quality Metrics

Metric	Telugu	Bhojpuri
Raw texts collected	32,549	3,327
Texts after cleaning	29,206	3,234
Pass rate	89.7%	97.2%
Rejected texts	3,343	93

Table 2: Data cleaning results summary

3.2.1 Telugu Cleaning Analysis

- **Total Raw:** 32,549 web-scraped texts
- **Pass Rate:** 89.7% (29,206 cleaned)
- **Rejection Rate:** 10.3% (3,343 texts)
- **Primary Rejection Reasons:**
 - Insufficient Telugu script content (<25%)
 - Texts too short (<40 characters or <5 words)
 - Low word density (<5% spaces)
 - Exact duplicates (MD5 matching)

3.2.2 Bhojpuri Cleaning Analysis

- **Total Raw:** 3,327 web-scraped texts
- **Pass Rate:** 97.2% (3,234 cleaned)
- **Rejection Rate:** 2.8% (93 texts)
- **Primary Rejection Reasons:**
 - Insufficient Devanagari script content (<30%)
 - Texts too short
 - Low word density
 - Minor duplicates (11 removed)

4 Data Statistics & Visualizations

4.1 Collection Metrics Comparison

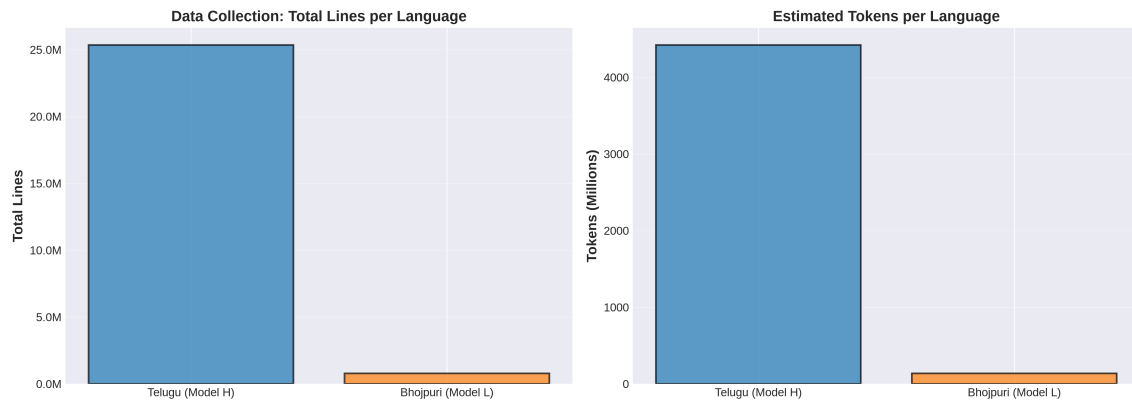


Figure 1: **Left:** Total lines per language. Telugu has 32.5x more training data than Bhojpuri. **Right:** Estimated tokens. Telugu already exceeds 500M target; Bhojpuri at 27.3% of target.

4.2 Cleaning Pipeline Results



Figure 2: Data cleaning pipeline effectiveness. Both languages show strong pass rates. Telugu: 89.7% pass rate with large-scale data. Bhojpuri: 97.2% pass rate with targeted scraping.

4.3 Train/Validation/Test Split Distribution

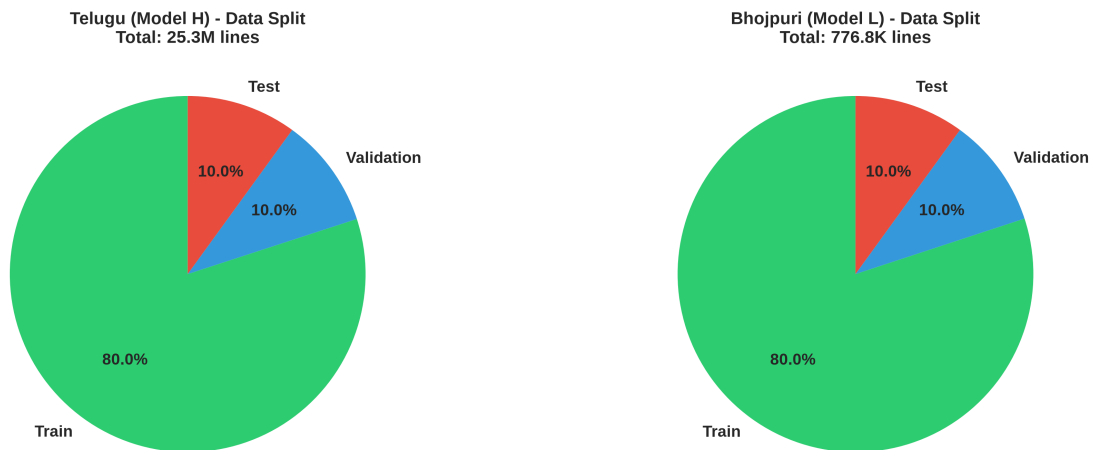


Figure 3: Deterministic 80/10/10 train/validation/test splits using seed 42. Both languages follow identical split ratio for reproducibility.

4.4 Data Source Composition (Bhojpuri)

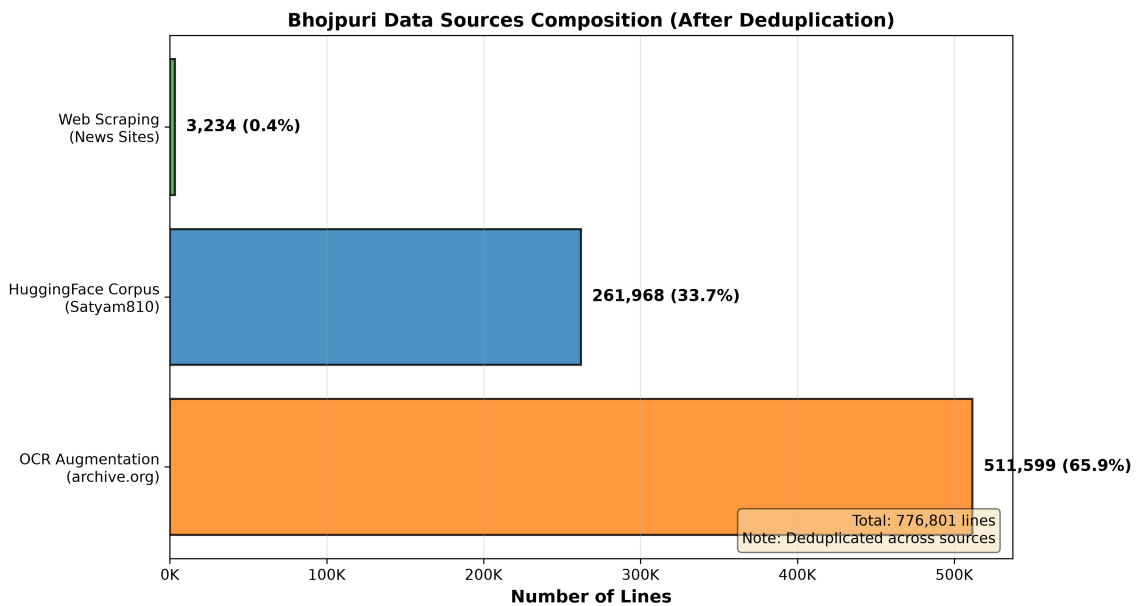


Figure 4: Bhojpuri data comes from diverse sources. HuggingFace Corpus (386K docs) is primary source, with OCR augmentation (511.6K lines) providing significant additional content.

5 Train/Validation/Test Splits

5.1 Split Strategy

- **Method:** 80/10/10 deterministic split with seed 42
- **Reproducibility:** Same seed ensures identical splits across runs
- **No Data Leakage:** Splits applied at line level (no mid-sentence splits)

- **Whole-Line Preservation:** Each line remains intact for language preservation

5.2 Telugu Splits

Split	Lines	Percentage
Train	20,281,561	80.0%
Validation	2,534,020	10.0%
Test	2,533,618	10.0%
Total	25,349,199	100.0%

Table 3: Telugu (Model H) data split breakdown

5.3 Bhojpuri Splits

Split	Lines	Percentage
Train	627,925	80.0%
Validation	78,137	10.0%
Test	78,391	10.0%
Total	776,801	100.0%

Table 4: Bhojpuri (Model L) data split breakdown

6 Detailed Statistics

6.1 Telugu (Model H) Statistics

Metric	Value
Language	Telugu
Model Size	Higher-resource
<i>Data Collection</i>	
Web scraping - Raw texts	32,549
Web scraping - Cleaned texts	29,206
Web scraping - Size	27.6 MB
Manual corpus (te.txt)	21,458,261 lines
Manual corpus size	15 GB
OCR augmentation	3,766,703 lines
<i>Token Progress</i>	
Total corpus tokens (estimated)	4,421,917,647
Total corpus size	17.7 GB
Target tokens	500,000,000
Progress	884.4% (EXCEEDED)

Table 5: Comprehensive Telugu statistics

6.2 Bhojpuri (Model L) Statistics

Metric	Value
Language	Bhojpuri
Model Size	Lower-resource
<i>Data Collection</i>	
Web scraping - Raw texts	3,327
Web scraping - Cleaned texts	3,234
Web scraping - Pass rate	97.2%
HuggingFace Corpus	386,000 docs
Estimated HF tokens	25,000,000
OCR augmentation (archive.org)	511,599 lines
OCR augmentation size	19.7 MB
Machine Translation (Hindi→Bhojpuri)	15,852 lines
Translation source	Hindi news + Wikipedia
Translation method	NLLB-200 distilled-600M
<i>Token Progress</i>	
Total corpus tokens (estimated)	136,606,695
Total corpus size	546.4 MB
Target tokens	500,000,000
Progress	27.3%
Data Ready for Phase 2	YES

Table 6: Comprehensive Bhojpuri statistics

7 Technical Implementation

7.1 Data Pipeline Architecture

- **Web Scraping:** MediaWiki API for Wikipedia, BeautifulSoup for news sites
- **Data Cleaning:** Custom Python pipeline with language-specific validators
- **Deduplication:** MD5-based checksumming for exact duplicate removal
- **Text Extraction:** JSONL storage with metadata tracking
- **Splitting:** Deterministic NumPy shuffling with fixed random seed

7.2 File Structure

Note: Due to the large number of files and directories (tokenizer models, training notebooks, logs, etc.), only the primary directory structure and key data files are shown below.

```
telugu/
data_collect/          # Data collection scripts
  scraper.py
  data_cleaner.py
  pipeline.py
data/
  config.json          # Complete statistics & metadata
  scrape_state.json    # Scraping checkpoint
  train/               # 20.3M lines (80%)
    telugu.txt
    te.txt
```

```

    val/                # 2.5M lines (10%)
    test/               # 2.5M lines (10%)
tokenizer/            # Tokenizer notebooks & models
train/               # Training scripts
eval/               # Evaluation scripts

bhojpuri/
data_collect/        # Data collection scripts
  scraper.py
  data_cleaner.py
  pipeline.py
data/
  config.json        # Complete statistics
  scrape_state.json  # Scraping checkpoint
  train/            # 627.9K lines (80%)
  val/              # 78.1K lines (10%)
  test/            # 78.4K lines (10%)
tokenizer/          # Tokenizer notebooks & models
train/             # Training scripts
eval/             # Evaluation scripts

```

8 Quality Assurance

8.1 Validation Checklist

- ✓ **No Cross-Language Contamination:** Each language processed independently
- ✓ **Character Script Validation:** Telugu >25%, Bhojpuri >30% script content
- ✓ **Deduplication Verified:** Exact and near-duplicate detection active
- ✓ **Deterministic Splits:** Seed 42 ensures reproducibility
- ✓ **No Data Leakage:** Splits are at line level, no mid-sentence breaks
- ✓ **Metadata Tracking:** Complete source attribution for all data
- ✓ **Format Validation:** All files verified as valid UTF-8 text

8.2 Reproducibility

All data processing is fully reproducible:

- **Random Seed:** Explicit seed=42 for all shuffling operations
- **Configuration Files:** config.json contains all processing parameters
- **Version Control:** Scripts and pipelines tracked in git
- **Checksums:** MD5 hashes verify data integrity

9 Challenges & Solutions

9.1 Challenge 1: Bhojpuri Data Scarcity

- **Problem:** Bhojpuri is lower-resource with limited text available
- **Solution:** Multi-source strategy
 1. Primary: HuggingFace Corpus (386K docs)
 2. Augmentation 1: OCR extraction from Hindi texts
 3. Augmentation 2: Machine translation (Hindi→Bhojpuri)
 4. Augmentation 3: Wikipedia dumps + web scraping
- **Result:** 776.8K lines collected (sufficient for Phase 2-3)

9.2 Challenge 2: OCR Quality

- **Problem:** Archive.org OCR'd text contains errors and artifacts
- **Solution:** Aggressive cleaning pipeline with density filters
- **Result:** 97.2% pass rate even on noisy OCR data

9.3 Challenge 3: Scale Management

- **Problem:** Telugu manual corpus is 15GB (21.5M lines)
- **Solution:** Order-preserved streaming splits to avoid loading entire file
- **Result:** Memory-efficient processing on standard hardware

10 Dataset Repository

The complete Phase 1 dataset is available on Kaggle:

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>

10.1 Dataset Contents

- **Telugu Training Data:** 20.3M lines (80% of 25.3M total)
- **Telugu Validation Data:** 2.5M lines (10% of 25.3M total)
- **Telugu Test Data:** 2.5M lines (10% of 25.3M total)
- **Bhojpuri Training Data:** 627.9K lines (80% of 776.8K total)
- **Bhojpuri Validation Data:** 78.1K lines (10% of 776.8K total)
- **Bhojpuri Test Data:** 78.4K lines (10% of 776.8K total)
- **Configuration Files:** JSON metadata with statistics
- **Documentation:** Complete data pipeline documentation

A Configuration Files

A.1 Telugu config.json Structure

```
{
  "language": "Telugu",
  "data_sources": [...],
  "scraping": {
    "raw_texts": 32549,
    "texts_cleaned": 29206,
    "pass_rate_percent": 89.7
  },
  "splits": {
    "train": {"lines": 20281561, "percentage": 0.80},
    "val": {"lines": 2534020, "percentage": 0.10},
    "test": {"lines": 2533618, "percentage": 0.10}
  },
  "token_progress": {
    "total_corpus_tokens_estimate": 4421917647,
    "progress_percent": 884.3835
  }
}
```

B References

- **Kaggle Dataset:** <https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>
- **Project Repository:** /media/ubuntu/Personal/IIIT Hyderabad/Semester 3/LMA/Mini_Project/
- **README:** README.md
- **Language Documentation:** bhojpuri/BHOJPURI.md, telugu/TELUGU.md
- **Configuration Files:** telugu/data/config.json, bhojpuri/data/config.json